SIMATS ENGINEERING ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE OUTCOME COVERED

COURSE CODE: CSA15

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) Question Count: 5

Code of Conduct

Total Marks: 25

I _____**Shaik Mohammad Akhim(192525132)**_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: __**Shaik Mohammad akhim**_____

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques.

Explain how load balancing and auto-scaling can ensure smooth streaming.

Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies.

Explain how service discovery and scaling can be handled efficiently.

Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements.

Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented.

Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down.

Propose a multi-region deployment strategy in the cloud.

Explain how data replication and failover mechanisms should be configured.

Discuss monitoring and alerting strategies for proactive issue detection.

ANSWERS:

1. Cloud Architecture for Video Streaming Platform

Scenario: A media company wants to provide video streaming services to millions of users without buffering.

1. Architecture using scalability and caching

- Use cloud storage to store videos.
- Use a Content Delivery Network (CDN) to deliver videos from nearby servers.
- Use caching to store frequently accessed videos.
- Use scalable cloud servers to handle millions of users.
- Use a database to manage user details and video information.

2. Load balancing and auto-scaling

- Load Balancing: Distributes user requests across multiple servers.
- Auto-scaling: Automatically increases or decreases the number of servers based on user traffic.
- During high traffic, more servers are added.
- During low traffic, unnecessary servers are removed.

This ensures smooth streaming and reduces buffering.

3. Cost optimization strategies

- Use pay-as-you-go pricing.
- Remove unused resources.
- Use auto-scaling to avoid unnecessary server costs.
- Store old videos in low-cost storage.

- Use CDN caching to reduce bandwidth costs.

Conclusion: Cloud scalability, caching, load balancing and auto-scaling provide smooth video streaming at a lower cost.

2. Container Orchestration and Multi-Cloud Strategy

Scenario: A fintech startup uses containers across multiple cloud providers and faces deployment problems.

1. Proposed solution

- Use Docker to create application containers.
- Use Kubernetes to manage containers.
- Deploy applications across multiple cloud providers.
- Use CI/CD pipelines for automatic application deployment.
- Maintain the same container configuration across all cloud environments.

2. Service discovery and scaling

- Service Discovery: Helps containers find and communicate with other services automatically.
- Auto-scaling: Kubernetes automatically increases or decreases container instances based on workload.
- Load balancing distributes requests between containers.
- Health checks identify failed containers and restart them. **3. Benefits and risks of multi-cloud deployment**

Benefits:

- Avoids dependency on a single cloud provider.
- Improves availability.
- Provides flexibility.
- Reduces service interruption.
- Helps select cost-effective cloud services.

Risks:

- Complex management.
- Higher maintenance costs.
- Security challenges.
- Network communication problems.
- Different cloud configurations.

Conclusion: Docker and Kubernetes provide consistent container deployment, efficient scaling and better application management across multiple clouds.

3. Cloud Backup and Disaster Recovery

Scenario: An e-learning platform loses important data due to accidental deletion.

1. Backup and disaster recovery solution

- Use automated cloud backups to protect important data.
- Enable data replication to another cloud region.
- Use versioning to recover previous versions of deleted files.
- Store backup copies in a separate location.
- Use disaster recovery services to restore applications and data.

2. RTO and RPO

RTO (Recovery Time Objective):

- Maximum acceptable time required to restore services after failure.
- Achieved using automated recovery and standby systems.

RPO (Recovery Point Objective):

- Maximum acceptable amount of data loss measured in time.
- Achieved using frequent backups and continuous data replication.

3. Business continuity steps

1. Identify the failure.
2. Activate the disaster recovery plan.
3. Restore data from backups.
4. Switch users to the backup system.
5. Verify data accuracy.
6. Resume normal operations.
7. Test the recovery system regularly.

Conclusion: Cloud backup, replication and versioning protect important data and help restore services quickly after failures.

4. Hybrid Cloud for Government Agency

Scenario: A government agency wants to store sensitive and non-sensitive data using a hybrid cloud model.

1. Hybrid cloud architecture

- Store sensitive data on private cloud or on-premises servers.
- Store non-sensitive data on public cloud.
- Connect both environments using a secure VPN or dedicated private connection.
- Use firewalls to control network traffic.
- Use access control to protect important information.

2. Identity management and encryption

- Use Identity and Access Management (IAM) to control user access.
- Apply Multi-Factor Authentication (MFA) for additional security.
- Use role-based access control (RBAC).
- Encrypt data during transmission using TLS.
- Encrypt stored data using strong encryption such as AES-256.
- Maintain regular security audits.

3. Challenges in hybrid cloud

- Complex management.
- High implementation cost.
- Data integration problems.
- Security risks.
- Network connectivity issues.
- Compliance and monitoring difficulties.

Conclusion: Hybrid cloud provides flexibility by keeping sensitive data on-premises while using public cloud resources for non-sensitive applications.

5. High Availability and Fault Tolerance in SaaS

Scenario: A SaaS provider wants its application to continue working even if one cloud region fails.

1. Multi-region deployment strategy

- Deploy applications in multiple cloud regions.
- Use load balancers to distribute user requests.
- Maintain backup servers in another region.
- Use auto-scaling to handle changing workloads.
- Deploy databases across multiple regions.

2. Data replication and failover

- Replicate data between different regions.
- Maintain backup copies of important information.
- Use automatic failover to switch traffic to a healthy region.
- Use health checks to identify failed servers.
- Synchronize data regularly to reduce data loss.

3. Monitoring and alerting strategies

- Monitor CPU, memory and network usage.
- Track application performance.
- Monitor database health.
- Configure alerts for server failures.
- Use centralized logging to identify errors.
- Perform regular disaster recovery tests.

Conclusion: Multi-region deployment, data replication, automatic failover and monitoring ensure high availability and fault tolerance in SaaS applications.

Important Keywords for Exam

Topic	**Important keywords**
Video Streaming	CDN, Caching, Load Balancing, Auto-scaling
Container Orchestration	Docker, Kubernetes, Service Discovery
Disaster Recovery	Backup, Replication, Versioning, RTO, RPO
Hybrid Cloud	VPN, IAM, MFA, Encryption, RBAC
High Availability	Multi-region, Failover, Monitoring, Fault Tolerance

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario

		understanding of the scenario context			
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification Clarity	20% 15%	Decisions are well justified and technically strong Clear and wellorganized response	Reasonable justification Generally clear	Weak justification Somewhat unclear	No useful justification Poorly presented

SET 2:

1.A retail company collects terabytes of customer transaction data every day from online and offline stores. The management wants to analyze purchasing patterns in near real time to improve product recommendations and inventory management.

- Design a cloud-based big data architecture using distributed storage and processing frameworks. ● Explain how batch processing and stream processing can be integrated.
- Discuss how data lakes and data warehouses can be used together.
- Recommend techniques to ensure scalability and cost efficiency.

2.A healthcare organization is migrating patient records and medical imaging data to the cloud. The solution must comply with strict privacy regulations while allowing authorized doctors to access records securely from multiple locations.

● Design a secure cloud architecture that protects sensitive healthcare data.

● Explain how encryption, identity and access management (IAM), and key management should be implemented.

● Discuss auditing, logging, and compliance monitoring strategies.

● Recommend methods to prevent unauthorized access and data breaches.

3.An online ticket booking platform experiences sudden spikes in traffic whenever popular events are announced. The company wants to reduce infrastructure management while ensuring fast response times.

● Design a serverless cloud architecture for the application.

● Explain how event-driven computing can improve scalability.

● Discuss the role of managed databases, API gateways, and message queues. ● Evaluate the advantages and limitations of serverless computing for this workload.

4.A multinational company gathers data from sales, finance, manufacturing, and customer support systems. The organization requires a centralized analytics platform with high-quality and consistent data.

● Design a cloud-based ETL/ELT pipeline for integrating data from multiple sources. ● Explain how data quality, metadata management, and governance should be maintained. ● Discuss the role of workflow orchestration and scheduling tools.

● Recommend strategies for optimizing query performance and storage costs.

5.A logistics company wants to predict delivery delays using historical shipment data, weather information, and live GPS tracking. The analytics platform must support continuous model training and real-time predictions.

● Design a cloud-based machine learning and analytics architecture.

● Explain how big data technologies can support model training and inference. ● Discuss the use of cloud-native AI/ML services for model deployment and monitoring. ● Recommend strategies for handling large-scale data ingestion, feature engineering, and model updates.

ANSWERS:

1. Big Data Architecture for Retail Company

Scenario: A retail company collects large amounts of customer transaction data from online and offline stores. It wants to analyze purchasing patterns in near real time.

1. Cloud-based Big Data Architecture

Use cloud storage such as Amazon S3 to store large amounts of data.

Use Apache Hadoop for distributed data storage and processing.

Use Apache Spark for fast data processing and analysis.

Collect data from online and offline stores.

Analyze customer purchases to improve product recommendations and inventory management.

2. Batch Processing and Stream Processing

Batch Processing: Processes large amounts of stored data at scheduled intervals.

Stream Processing: Processes incoming data immediately as it arrives.

Use Apache Spark for batch and stream processing.

Combine both methods to analyze past and current customer purchases.

3. Data Lakes and Data Warehouses

Data Lake: Stores raw data in different formats such as images, files and transactions.

Data Warehouse: Stores cleaned and structured data for reporting and analysis.

Data lakes store large amounts of raw information.

Data warehouses help management make business decisions.

4. Scalability and Cost Efficiency

Use auto-scaling to increase or decrease resources.

Use distributed storage to handle large datasets.

Remove duplicate and unnecessary data.

Use low-cost storage for old data.

Use pay-as-you-go cloud services.

Conclusion: Big data technologies help retail companies analyze customer behavior, improve recommendations and manage inventory efficiently.

2. Secure Cloud Architecture for Healthcare Organization

Scenario: A healthcare organization wants to store patient records and medical images in the cloud while protecting patient privacy.

1. Secure Cloud Architecture

Use a private cloud or secure cloud environment.

Store patient records in encrypted databases.

Store medical images in secure cloud storage.

Allow only authorized doctors to access patient information.

Use VPN for secure communication.

Maintain regular backups.

2. Encryption, IAM and Key Management

Encryption: Protects patient data from unauthorized access.

IAM: Controls who can access patient records.

MFA: Requires additional verification during login.

RBAC: Gives access based on user roles.

Key Management: Securely creates, stores and manages encryption keys.

Use encryption such as AES-256 for stored data and TLS for data transmission.

3. Auditing, Logging and Compliance Maintain records of every user activity.

Monitor login attempts and data access.

Generate security reports regularly.

Use alerts to identify suspicious activities.

Follow healthcare privacy regulations such as HIPAA where applicable.

4. Preventing Unauthorized Access

Use strong passwords and MFA.

Apply role-based access control.

Encrypt sensitive information.

Install firewalls and security monitoring tools.

Perform regular security audits.

Take regular backups.

Conclusion: Secure cloud architecture, encryption, IAM and monitoring protect patient information and provide safe access to authorized doctors.

3. Serverless Architecture for Online Ticket Booking

Scenario: An online ticket booking platform experiences sudden traffic increases when popular events are announced.

1. Serverless Cloud Architecture

Use AWS Lambda or Azure Functions to execute application code.

Use API Gateway to receive user requests.

Use a managed database to store booking details.

Use message queues to manage booking requests.

Use cloud storage to store event information.

2. Event-Driven Computing

Executes functions when an event occurs.

For example, when a user books a ticket, a function processes the request.

Automatically handles increasing user requests.

Runs functions only when required.

Reduces manual server management.

3. Role of Cloud Services

Managed Database: Stores user details, ticket information and booking records.

API Gateway: Connects users with backend services.

Message Queue: Stores booking requests and processes them in order.

Serverless Functions: Execute booking and payment

operations. **4. Advantages and Limitations** Advantages:

No need to manage servers.

Automatic scaling.

Pay only for actual usage.

Faster development.

Reduced infrastructure cost.

Limitations:

Cold start delays.

Execution time limits.

Difficult debugging.

Vendor dependency.

Database connection management challenges.

Conclusion: Serverless computing provides automatic scaling, reduced server management and efficient ticket booking during sudden traffic increases.

4. Cloud-Based ETL/ELT Pipeline for Multinational Company

Scenario: A multinational company collects data from sales, finance, manufacturing and customer support systems. It needs a centralized analytics platform.

1. Cloud-Based ETL/ELT Pipeline

Collect data from different departments.

Use ETL tools to extract, transform and load data.

Store collected data in cloud storage.

Clean and organize the data.

Load processed data into a cloud data warehouse.

Generate reports and business insights.

ETL: Extract → Transform → Load.

ELT: Extract → Load → Transform.

2. Data Quality, Metadata and Governance Remove duplicate and incorrect records.

Check missing values.

Maintain consistent data formats.

Use metadata to describe data.

Apply data governance policies.

Control data access using IAM.

3. Workflow Orchestration and Scheduling Use Apache Airflow to manage workflows.

Schedule data processing jobs.

Automatically execute tasks.

Monitor pipeline failures.

Retry failed tasks.

Maintain proper execution order.

4. Query Performance and Storage Optimization

Use partitioning to organize large datasets.

Use indexing to improve data searching.

Apply compression to reduce storage size.

Remove unnecessary data.

Use columnar formats such as Parquet.

Use auto-scaling for changing workloads.

Conclusion: A cloud-based ETL/ELT pipeline integrates data from different departments and provides accurate information for business decision-making.

5. Machine Learning Architecture for Logistics Company

Scenario: A logistics company wants to predict delivery delays using historical shipment data, weather information and live GPS tracking.

1. Cloud-Based Machine Learning Architecture

Collect historical shipment data.

Collect weather information and live GPS data.

Store data in cloud storage.

Use Apache Spark for large-scale data processing.

Train machine learning models using cloud ML services.

Predict delivery delays based on collected information.

Display predictions through a dashboard.

2. Big Data Technologies for Training and Inference

Use Hadoop for distributed data storage.

Use Apache Spark for large-scale data processing.

Use historical data to train machine learning models.

Use live GPS and weather data for real-time predictions.

Process incoming data to update delivery predictions.

3. Cloud-Native AI/ML Services

Use AWS SageMaker, Azure Machine Learning or Google Vertex AI.

Train machine learning models in the cloud.

Deploy models as prediction APIs.

Monitor model accuracy and performance.

Retrain models when new data becomes available.

4. Data Ingestion, Feature Engineering and Model Updates

Use streaming tools to collect live GPS data.

Store large datasets in distributed cloud storage.

Clean missing and incorrect data.

Extract useful features such as distance, weather and traffic conditions.

Train models using updated datasets.

Continuously monitor and update models.

Conclusion: Cloud-based machine learning and big data technologies help logistics companies predict delivery delays, improve transportation planning and provide better customer service.

Important Keywords for Revision

Question	Important keywords
1. Retail Big Data	Hadoop, Spark, Batch Processing, Stream Processing, Data Lake
2. Healthcare Security	Encryption, IAM, MFA, AES-256, HIPAA, Auditing
3. Serverless Computing	AWS Lambda, API Gateway, Message Queue, Managed Database
4. ETL/ELT Pipeline	Extract, Transform, Load, Airflow, Metadata, Governance
5. Machine Learning	Hadoop, Spark, SageMaker, Feature Engineering, Model Training

SET 3 :

1. A manufacturing company plans to migrate its legacy enterprise applications and databases from an on-premises data center to the cloud. The organization wants to minimize downtime while ensuring application performance and security.

- Design a cloud migration strategy suitable for the organization.
- Explain how different migration approaches (Rehost, Refactor, Replatform, Repurchase, Retire, Retain) can be applied.
- Discuss methods to validate application performance after migration.
- Recommend strategies to reduce migration risks and downtime.

2. A smart city project deploys thousands of IoT sensors to monitor traffic, air quality, energy consumption, and public safety. The collected data must be processed and analyzed in real time for decision-making.

- Design a cloud-based architecture for IoT data collection and analytics.
- Explain how message brokers, stream processing, and cloud storage are integrated. ● Discuss methods for handling high-volume, high-velocity sensor data.
- Recommend techniques to ensure scalability, reliability, and security.

3. A global e-commerce company wants executives to monitor sales performance, customer behavior, and regional trends through interactive dashboards. The solution must support realtime analytics and historical reporting.

- Design a cloud-based business intelligence architecture.
- Explain how data visualization tools integrate with cloud data warehouses.
- Discuss techniques for optimizing dashboard performance with large datasets. ● Recommend security measures to protect business reports and sensitive data.

4. An online gaming company experiences fluctuating workloads throughout the day. During peak hours, users report high latency and reduced application performance.

- Design a cloud monitoring and performance optimization solution.

● Explain how cloud monitoring tools, metrics, logs, and tracing help identify performance bottlenecks.

● Discuss strategies for resource optimization and capacity planning.

● Recommend methods to improve application availability while controlling operational costs.

5. A financial institution stores petabytes of customer transaction data for regulatory compliance and business analytics. The organization must retain historical records while managing storage costs efficiently.

● Design a cloud-based data lifecycle management strategy.

● Explain how data classification, retention policies, and archival storage should be implemented. ● Discuss the importance of governance, auditing, and regulatory compliance in big data environments.

● Recommend techniques for balancing data accessibility, security, and cost optimization.

ANSWERS:

1. Cloud Migration Strategy for Manufacturing Company

Scenario: A manufacturing company wants to move its existing applications and databases from an on-premises data center to the cloud with minimum downtime.

1. Cloud Migration Strategy

First, analyze existing applications and databases.

Select a suitable cloud provider such as AWS, Azure or Google Cloud.

Create a backup of all important data.

Migrate applications and databases in phases.

Test the applications after migration.

Monitor performance and security.

2. Six Migration Approaches

1. Rehost: Move applications to the cloud without changing their code.
2. Refactor: Modify applications to use cloud-native technologies.
3. Replatform: Make small changes to improve application performance in the cloud.
4. Repurchase: Replace existing applications with cloud-based software.
5. Retire: Remove applications that are no longer required.
6. Retain: Keep some applications on-premises because of security or business requirements.

3. Validate Application Performance

Test application speed and response time.

Check CPU and memory usage.

Test database performance.

Perform load testing.

Monitor application errors.

Compare performance before and after migration.

4. Reduce Migration Risks and Downtime

Take regular backups.

Migrate applications step by step.

Use data replication.

Perform testing before migration.

Maintain a rollback plan.

Use secure network connections.

Conclusion: A proper cloud migration strategy reduces downtime, improves application performance and provides better security.

2. Cloud-Based IoT Architecture for Smart City

Scenario: A smart city uses thousands of IoT sensors to monitor traffic, air quality, energy and public safety.

1. Cloud-Based IoT Architecture

Install IoT sensors to collect real-time data.

Use IoT gateways to connect sensors to the cloud.

Use message brokers such as MQTT or Apache Kafka to transfer data.

Use cloud processing services to analyze data.

Store collected data in cloud storage.

Display results through dashboards.

2. Message Brokers, Stream Processing and Cloud Storage

Message Broker: Transfers sensor data between devices and cloud applications.

Stream Processing: Analyzes incoming data immediately.

Cloud Storage: Stores sensor data for future analysis.

Example: Traffic sensor data is processed to identify traffic congestion.

3. Handling High-Volume Sensor Data

Use distributed storage.

Apply data compression.

Use Apache Kafka for high-speed data collection.

Use Apache Spark for large-scale processing.

Remove duplicate and unnecessary data.

Process data in parallel.

4. Scalability, Reliability and Security

Use auto-scaling to handle increasing sensor data.

Use load balancing to distribute workloads.

Maintain backup copies of data.

Use encryption for data security.

Apply authentication to IoT devices.

Monitor sensor and cloud system performance.

Conclusion: Cloud-based IoT architecture helps smart cities collect and analyze real-time data for better traffic management, public safety and energy management.

3. Cloud-Based Business Intelligence Architecture

Scenario: A global e-commerce company wants executives to monitor sales, customer behavior and regional performance using interactive dashboards.

1. Cloud-Based Business Intelligence Architecture Collect data from sales, customers and regional stores.

Use ETL tools to clean and transform data.

Store processed data in a cloud data warehouse.

Connect BI tools to the data warehouse.

Create interactive dashboards and reports.

Help executives make better business decisions.

2. Data Visualization and Cloud Data Warehouses

Use Power BI, Tableau or Looker for visualization.

Connect visualization tools to cloud data warehouses.

Display sales reports using charts and graphs.

Show customer behavior and regional trends.

Provide real-time business information.

3. Dashboard Performance Optimization

Use data filtering to reduce unnecessary records.

Apply indexing and partitioning.

Use caching for frequently accessed reports.

Use data aggregation.

Optimize database queries.

Use scalable cloud resources.

4. Security Measures

Use IAM to control user access.

Apply role-based access control.

Encrypt sensitive business data.

Enable multi-factor authentication.

Maintain audit logs.

Restrict access to confidential reports.

Conclusion: Cloud-based business intelligence provides real-time analytics, interactive dashboards and useful business insights for decision-making.

4. Cloud Monitoring and Performance Optimization for Online Gaming

Scenario: An online gaming company experiences high latency and poor performance during peak hours.

1. Cloud Monitoring and Performance Optimization

Use cloud monitoring tools to track application performance.

Use load balancing to distribute player requests.

Use auto-scaling to add servers during peak hours.

Use CDN to deliver game content faster.

Monitor server and database performance.

Optimize application code.

2. Monitoring Tools, Metrics, Logs and Tracing

Monitoring Tools: AWS CloudWatch, Azure Monitor and Google Cloud Monitoring.

Metrics: Measure CPU, memory, network usage and response time.

Logs: Record application errors and user activities.

Tracing: Tracks requests across different services to identify delays.

These tools help identify performance problems and improve application speed.

3. Resource Optimization and Capacity Planning

Use auto-scaling.

Monitor CPU and memory usage.

Remove unused cloud resources.

Allocate resources based on demand.

Use load balancing.

Predict future traffic using historical data.

4. Improve Availability and Control Costs Deploy applications across multiple regions.

Use backup servers.

Configure automatic failover.

Use reserved instances for predictable workloads.

Use pay-as-you-go services.

Monitor cloud spending regularly.

Conclusion: Cloud monitoring, auto-scaling and load balancing improve gaming performance, reduce latency and control operational costs.

5. Cloud-Based Data Lifecycle Management for Financial Institution

Scenario: A financial institution stores large amounts of customer transaction data for regulatory compliance and business analysis.

1. Cloud-Based Data Lifecycle Management Collect transaction data from banking systems.

Classify data based on importance and sensitivity.

Store frequently accessed data in high-performance storage.

Move old data to low-cost archival storage.

Apply retention policies.

Delete data securely after the permitted retention period.

2. Data Classification, Retention and Archival Storage

Data Classification: Divide data into public, confidential and sensitive categories.

Retention Policy: Define how long data must be stored.

Archival Storage: Store old and rarely accessed data at a lower cost.

Use services such as Amazon S3 Glacier for long-term storage.

Maintain backups for important transaction records.

3. Governance, Auditing and Regulatory Compliance

Follow financial data protection regulations.

Maintain proper data governance policies.

Record all user activities.

Perform regular security audits.

Control access using IAM.

Maintain accurate transaction records.

Monitor suspicious activities.

4. Balance Accessibility, Security and Cost

Use different storage classes based on data usage.

Encrypt stored and transmitted data.

Apply role-based access control.

Use data compression.

Remove duplicate data.

Automate data movement between storage classes.

Maintain regular backups.

Conclusion: Cloud-based data lifecycle management helps financial institutions securely store historical records, follow regulatory requirements and reduce storage costs.

Important Keywords for Revision

Question	Important keywords
1. Cloud Migration	Rehost, Refactor, Replatform, Repurchase, Retire, Retain
2. Smart City IoT	IoT Sensors, MQTT, Kafka, Stream Processing, Cloud Storage
3. Business Intelligence	Data Warehouse, Power BI, Tableau, ETL, Dashboard
4. Gaming Performance	CloudWatch, Metrics, Logs, Tracing, Auto-scaling
5. Data Lifecycle	Data Classification, Retention, Archival, Governance, Auditing